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# ITECH3102 – Data Visualization
# Week 8

## Lab Exercise – Complex data visualizations (3)

**ID: _____Name: _____**

In this week's projects, we apply Tableau's built-in analytics tools to tackle analytical questions, which cannot be answered with simple descriptive visualizations. For example, we may want to see the trend or make predictions for future values. Tableau's built-in features such as trend models, clustering, distributions, and forecasting allow us to quickly add the output of analytics tools to the charts we create.

Tableau's advanced analytics tools can be found in the **Analytics pane**, which sits behind the Data pane on the left side of the screen. The tools are grouped as follows:

**Summary:**

- **Constant Line:** Adds a line to the chart, intersecting the axis at a fixed value of your choosing.
- **Average Line:** Adds one or several average lines to your view, based on the arithmetic mean of the chart's data.
- **Median with Quartiles:** Adds one or more sets of median lines with distribution bands to your view.

- **Box Plot**: Adds one or more box plots to your view.
- **Totals:** Adds subtotals, column grand totals, or row grand totals to your view.

**Model:**

- **Average with 95% CI**: Adds one or more sets of average lines with distribution bands to your view. The bands show the 95% confidence interval around the data's mean.
- **Median with 95% CI:** Adds one or more sets of median lines with distribution bands to your view. The bands show the 95% confidence interval around the data's median.
- **Trend Line:** Adds one or more trend lines to your view.
- **Forecast:** Adds a forecast to your view
- **Cluster:** Runs a clustering algorithm

**Custom:**

- **Reference Line:** Adds one or more lines reflecting the value of additional measures, calculations, or parameters.
- **Reference Band:** Adds one or more bands reflecting the values of additional measures, calculations, or parameters.
- **Distribution Band:** Adds a shaded band showing the distribution of the chart's data along an axis, as measured by percentages, percentiles, quantiles, or the standard deviation.
- **Box Plot:** Adds one or more customizable box plots to your view.

## Project 1: Visualize trend

Line charts can sometimes be difficult to interpret when the data is very granular. Trend lines help you to see the pattern in your data, by tracing out the fundamental evolution of the measure in question.

The following trend line models are available:

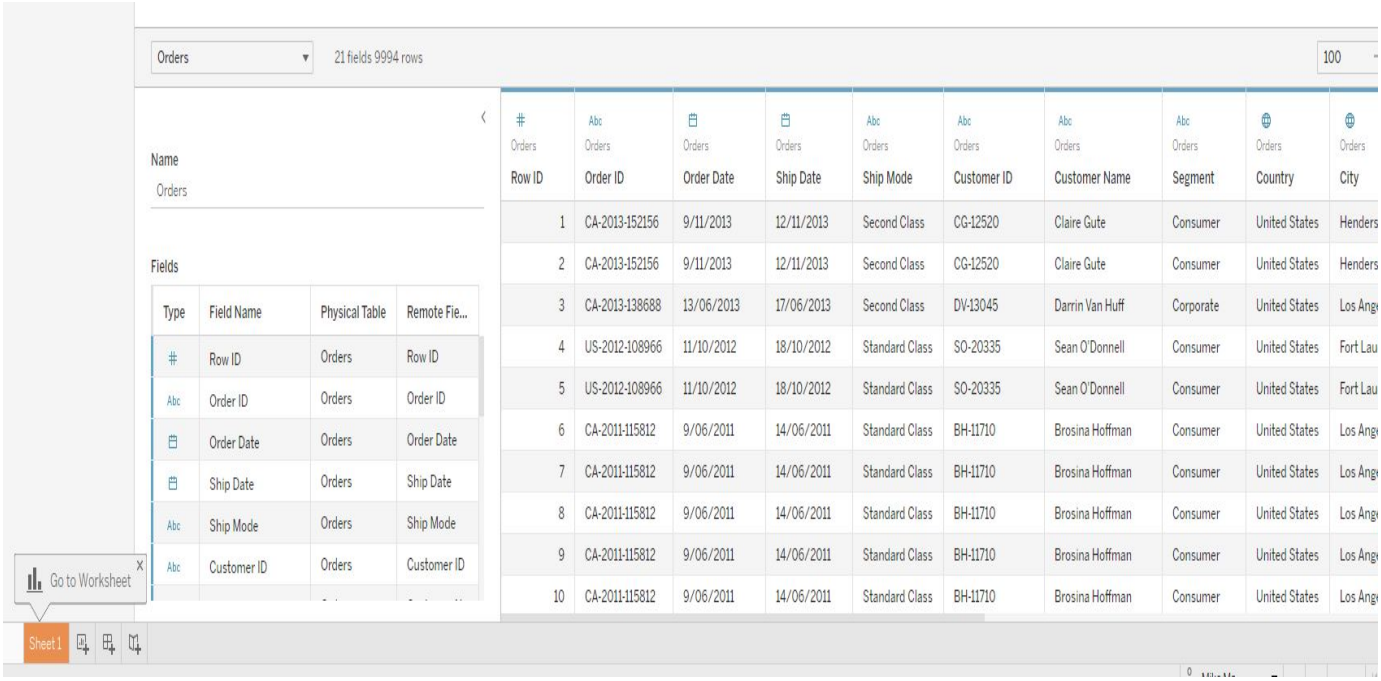
- Linear
- Logarithmic
- Exponential
- Polynomial
- Power

**Task:** The given dataset provides information about order, including sales, profit, city, order ID, etc. The goal of this task is to analyze the pattern and observe the relationship between the two variables in the dataset.

**1. Download Dataset:** order-information.xlsx from Week 8 in Moodle

## Step 1. Load the data into Tableau

**Drag** Orders table to canvas and drop it and check that your data has been loaded correctly.



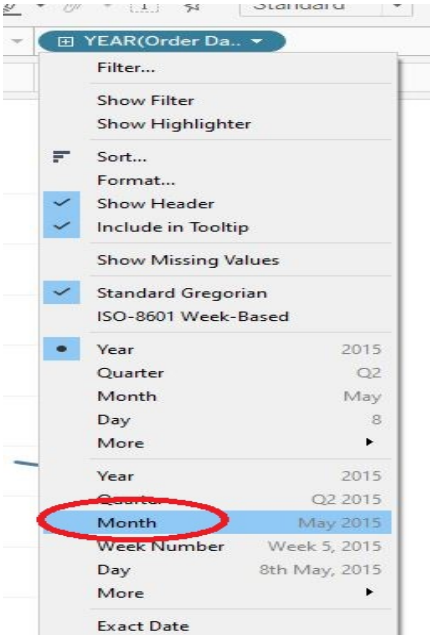

After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

## Step 2. Create a line chart

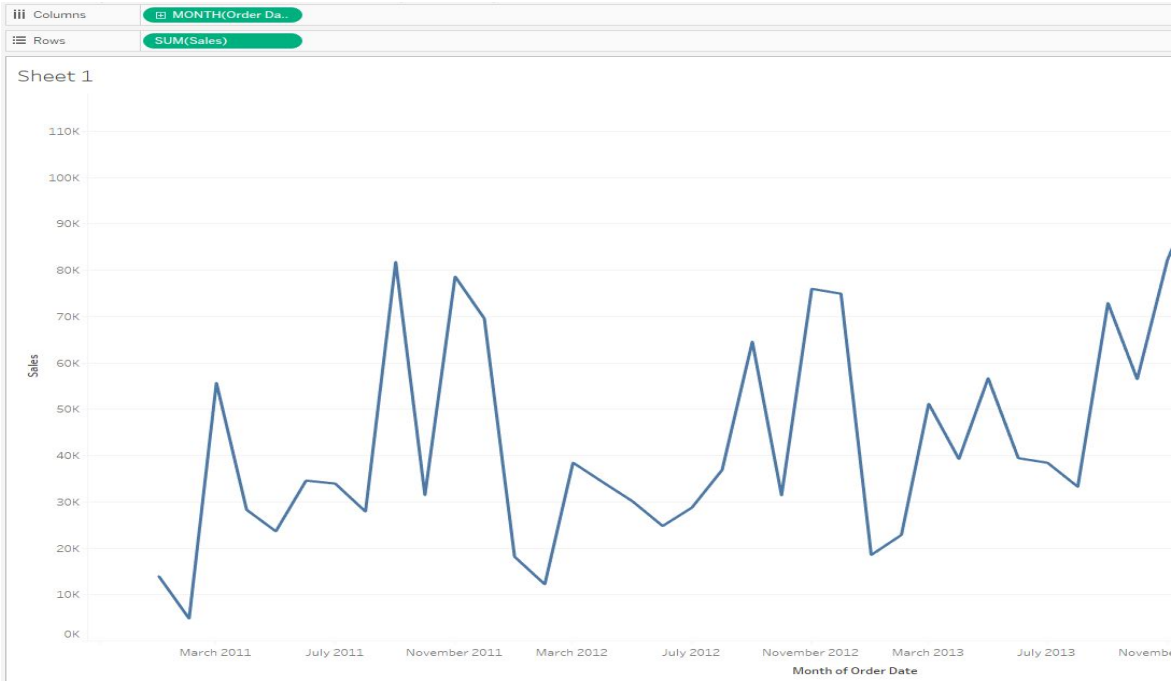
Drag the Sales field from the Measures area on Data pane to the Rows shelf, and drag the Order Date to the Columns shelf,




then click YEAR (Order Date) and select Month and Continuous:

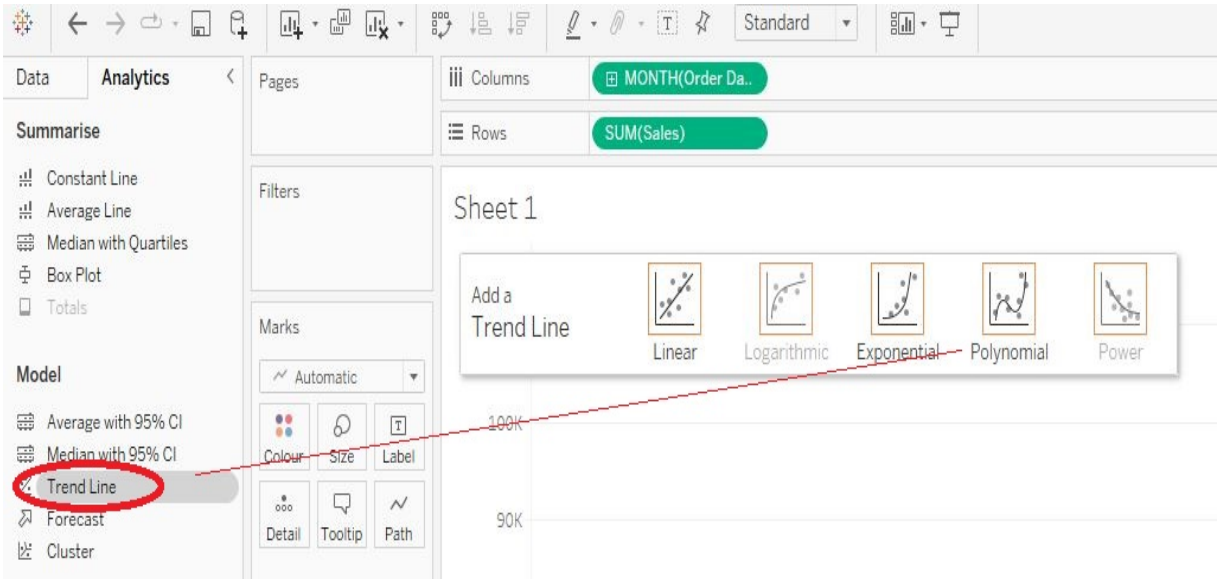


Now we get a line chart:

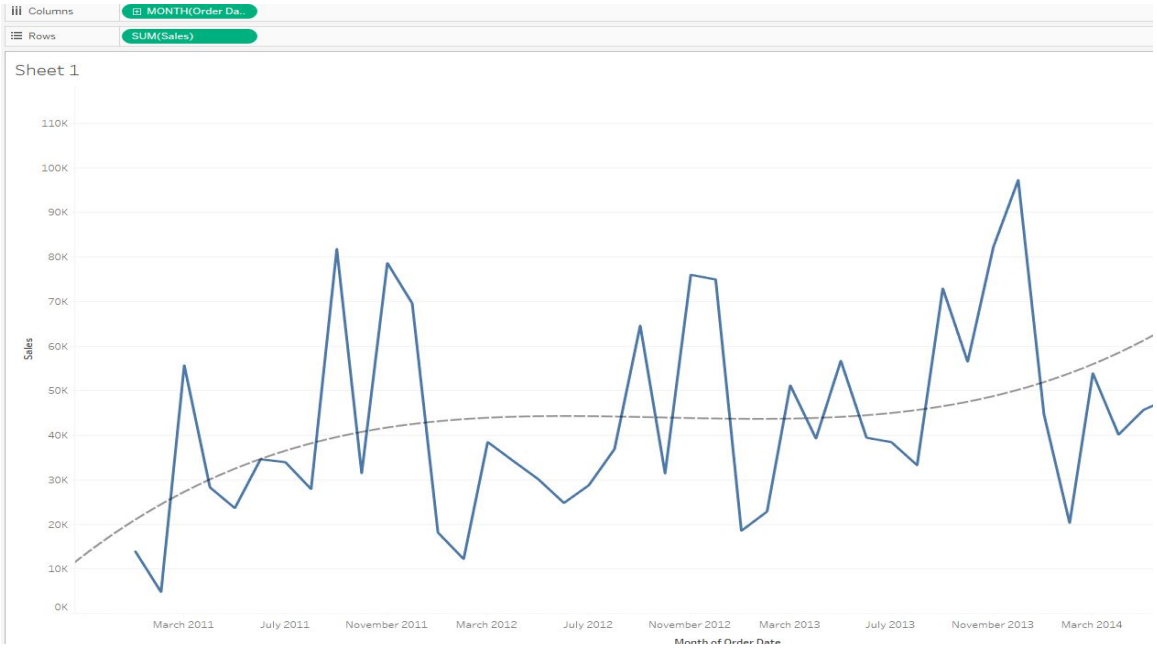



**Step 3. Add a trend line**

Click Analytics pane on left-hand and drag the Trend Line item from the Analytics pane to the canvas. While you are moving the mouse, Tableau will show a selection of the available types of regression models.



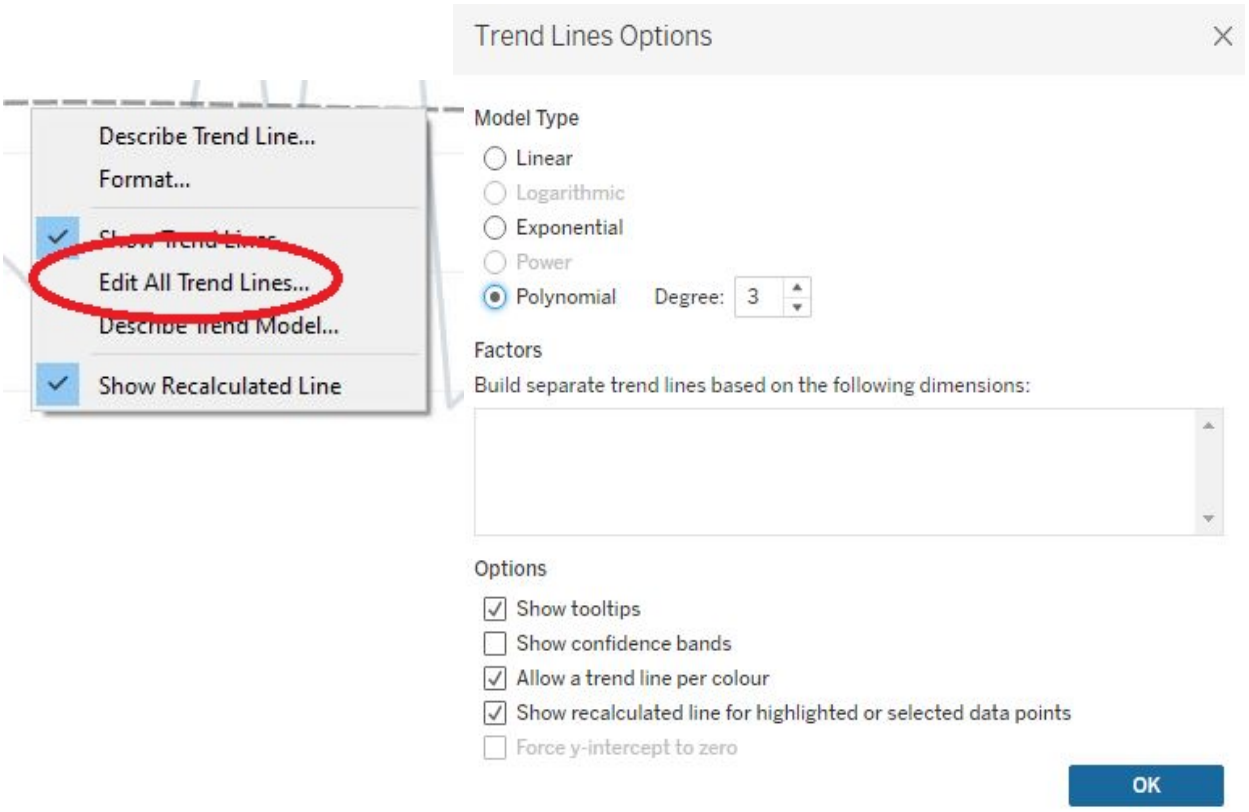

After dropping the Trend Line on the respective field (e.g., Polynomial), we get




From the trend line, we can see that revenues are trending up over the long run.

**Step 4. Configure a trend line**

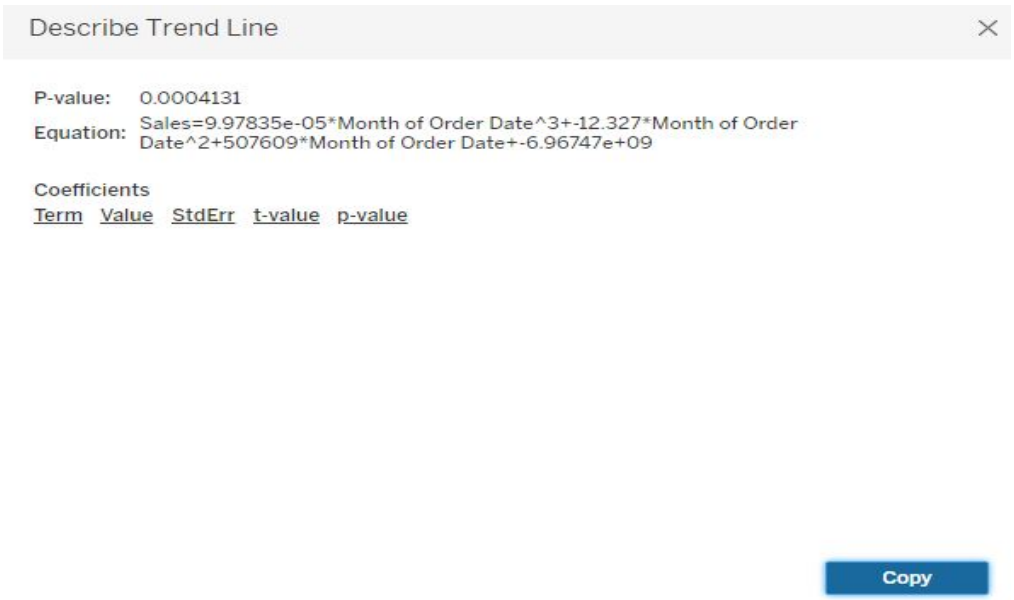
Right-click the trend line, and select Edit All Trend Lines. A dialog box with the options is shown



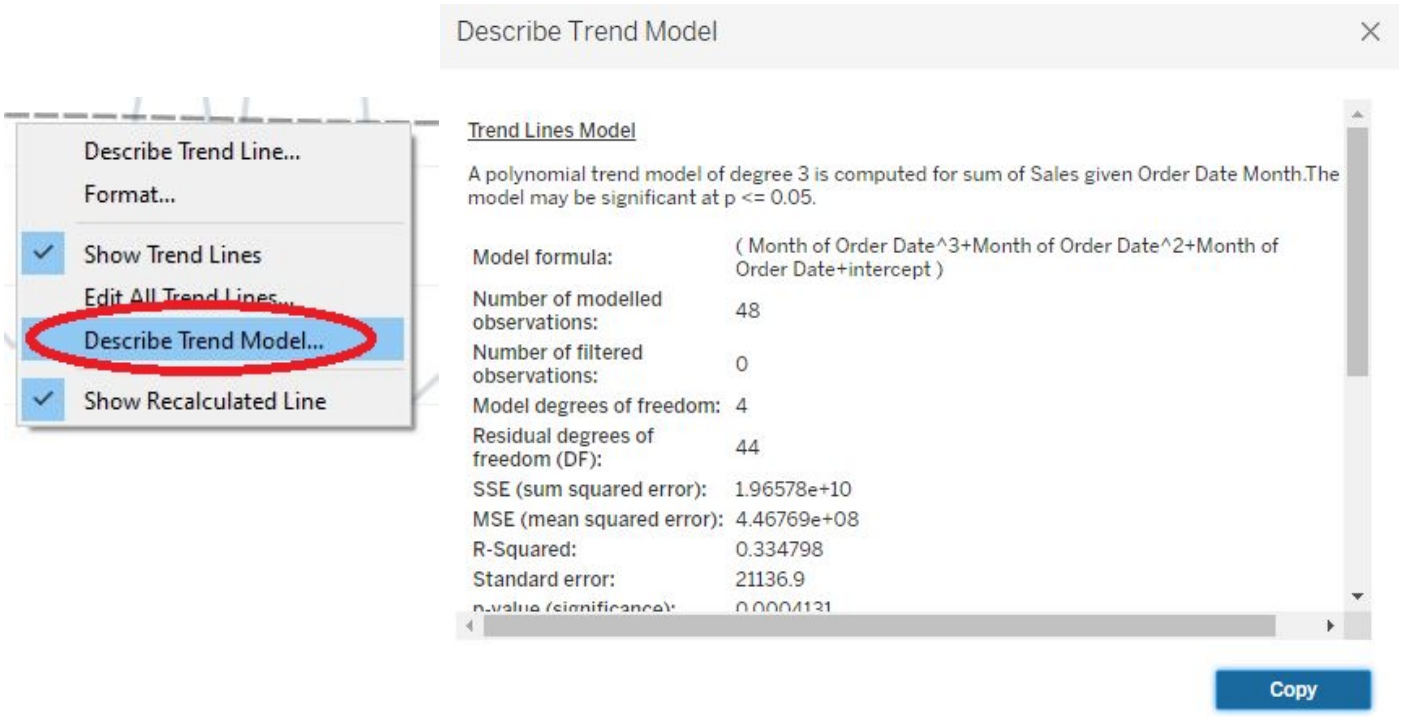

You can adjust the degree of polynomial trend lines. For example, if the option of Allow A Trend Line Per Color is checked, you can have a separate trend line for each group of marks of a certain color if your view includes a dimension on Color.

**Line and Trend Model Description**

Right- click the trend line and select Describe Trend Line. Tableau will show you the regression equation and the p-value for its overall significance, as shown




Right-click the trend line, and select Describe Trend Model. A dialog box with Describe Trend Model is shown




The trend model description shows the statistics of the model underlying your trend line.

## Project 2: Visualize forecasts

Forecasting models extrapolate future values of a time series based on its historical values, allowing you to attempt to predict the evolution of a measure.

Forecasting models work reasonably well to capture both long-term trends and any potential seasonality that may be present in the time series. The resulting forecast is shown directly in the chart.

Because we are talking about time-series data, ensure that your view contains a time or date field as well as a **measure**.

**Task:** The given dataset provides information about order, including sales, profit, city, order ID, etc. The goal of this task is to predict sales revenue for the next year using the order-information dataset.

**1. Download Dataset:** order-information.xlsx from Week 8 in Moodle

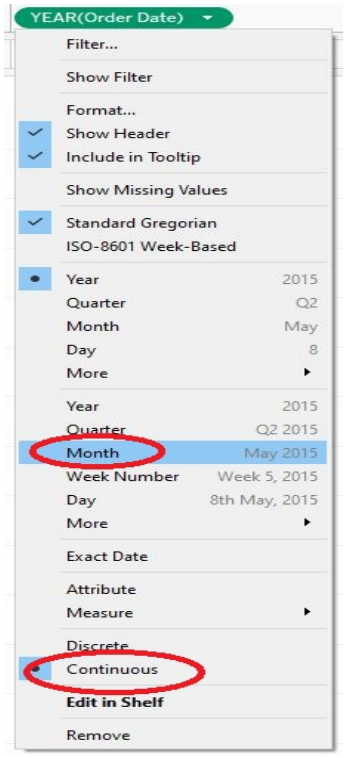
### Step 1. Load the data into Tableau

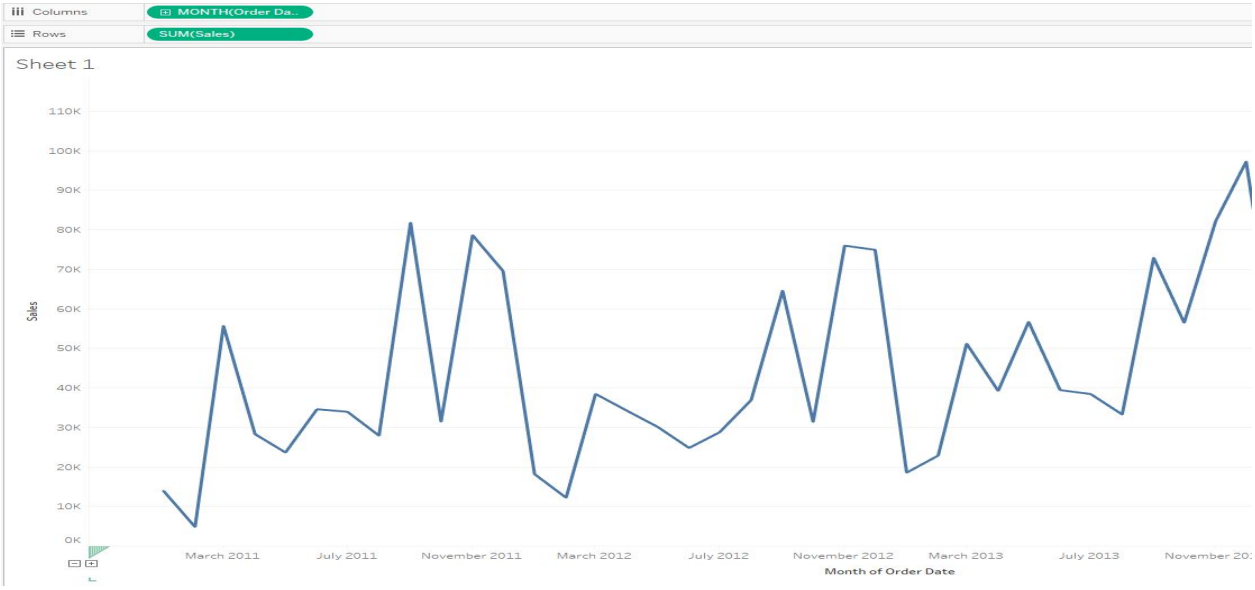
**Drag** Orders table to canvas and drop it and check that your data has been loaded correctly.

After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

### Step 2. Create a line chart

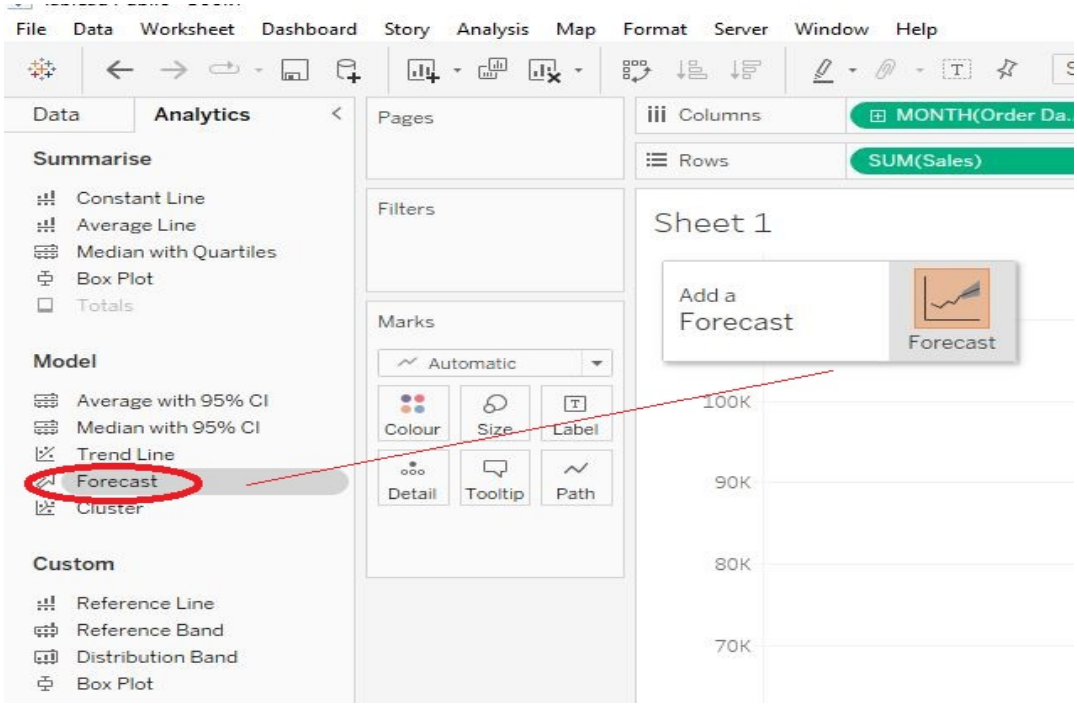
Drag the Sales field from the Measures area on Data pane to the Rows shelf, and drag the Order Date to the Columns shelf. Make sure that Month and Continuous are selected for YEAR (Order Date)

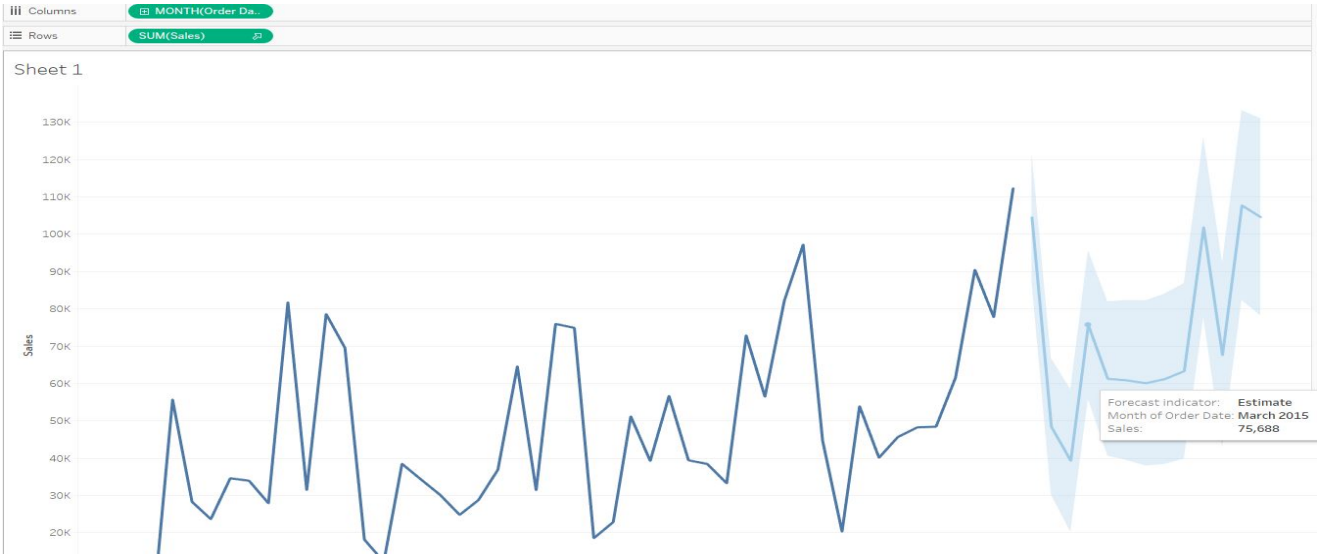



**Step 3. Add a forecast line to the view**

Click Analytics pane on left-hand and drag the Forecast item from the Analytics pane to the canvas. While you are moving the mouse, Tableau will show a Add a Forecast.

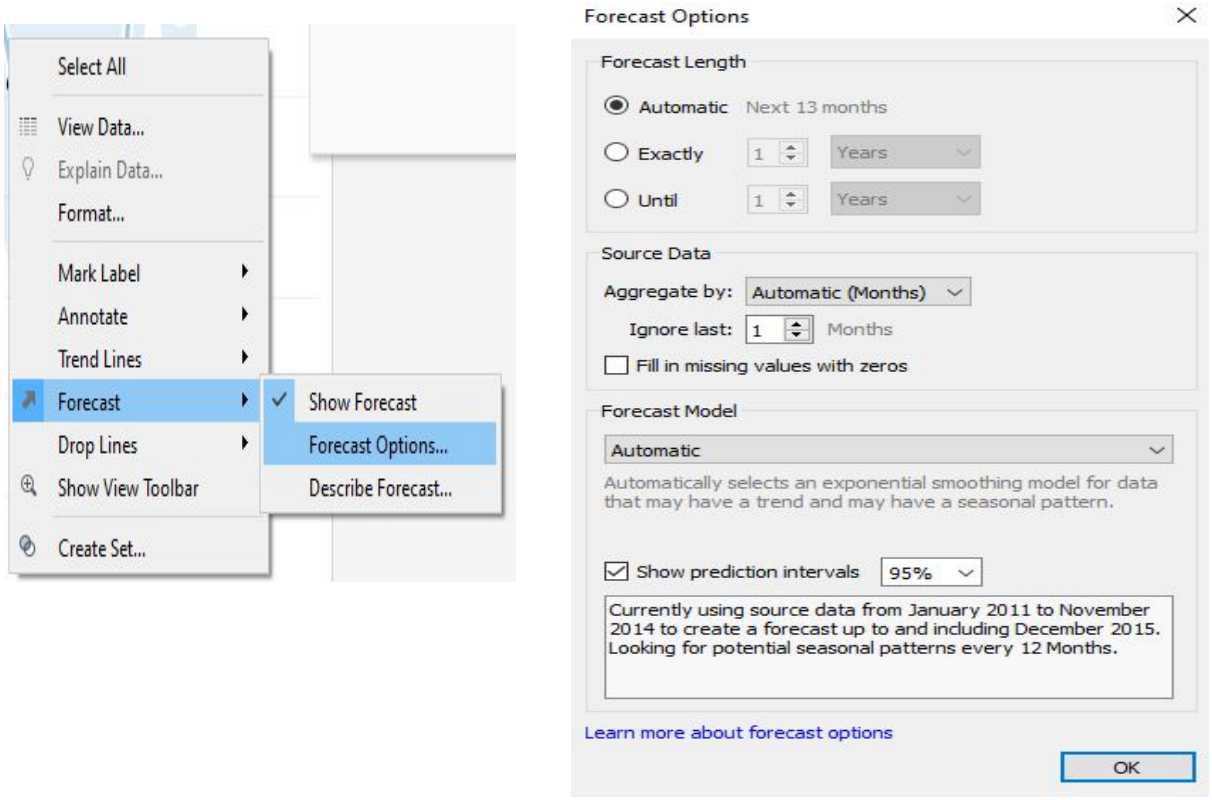



Your chart will be expanded with the forecasted values, which are shown in a lighter color.

**Step 4. Configure a forecast line**

Right-click anywhere on your chart and select Forecast | Forecast Options in the context menu. The Forecast Options window will appear.




From Forecast Options, you can see that, for this chart,

- Tableau suggests a forecast length of 13 months. This length was chosen based on the amount of historical data available, but you can change this value in accordance with your needs.
- The Ignore Last field allows you to leave out data for incomplete months that might skew the results.
- Missing values are automatically excluded from forecast models in Tableau. Depending on the situation, this may be the right thing to do. But there are cases where it can make sense to activate the option

## Project 3: Visualize cluster

A cluster is a collection of data points with similar properties. Cluster analysis helps you find such groupings in your data. A classic use case comes from the field of marketing, where clustering is often used to define different customer segments.

**Task:** The given dataset provides information about order, including sales, profit, city, order ID, etc. The goal of this task is to segment the customers in the order-information dataset using two variables: profitability and sales volume.

**1. Download Dataset:** order-information.xlsx from Week 8 in Moodle

### Step 1. Load the data into Tableau

**Drag** Orders table to canvas and drop it and check that your data has been loaded correctly.

After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

### Step 2. Create a scatter plot

Drag Profit field from the Measures area on Data pane to Rows shelf, drag the Sales field from the Measures area on Data pane to the Columns shelf, and drag Customer ID from the dimension area on Data pane to Detail on the Marks card in order to get one mark per customer.




We first create a new calculated field called RecydableTonsCollected. Right click any location in Dimensions or Measures area in Data pane and select Create | Create Calculated Field.

**Step 3. Create clusters**

From the Analytics pane, drag the Cluster item onto the view.

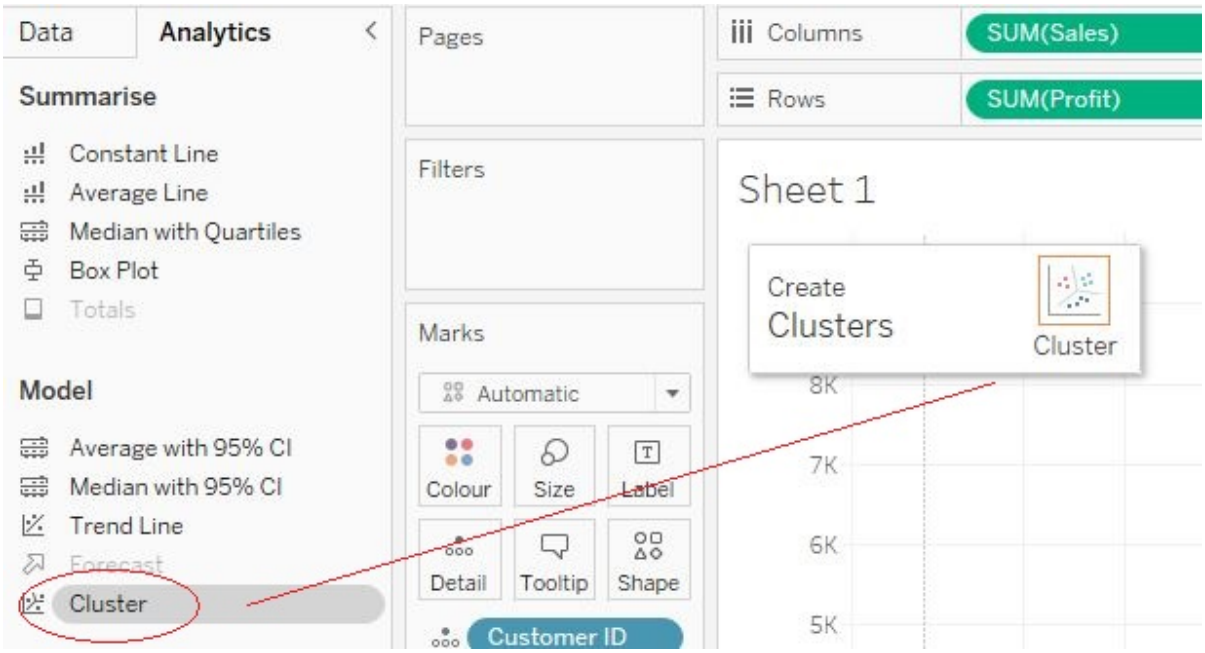
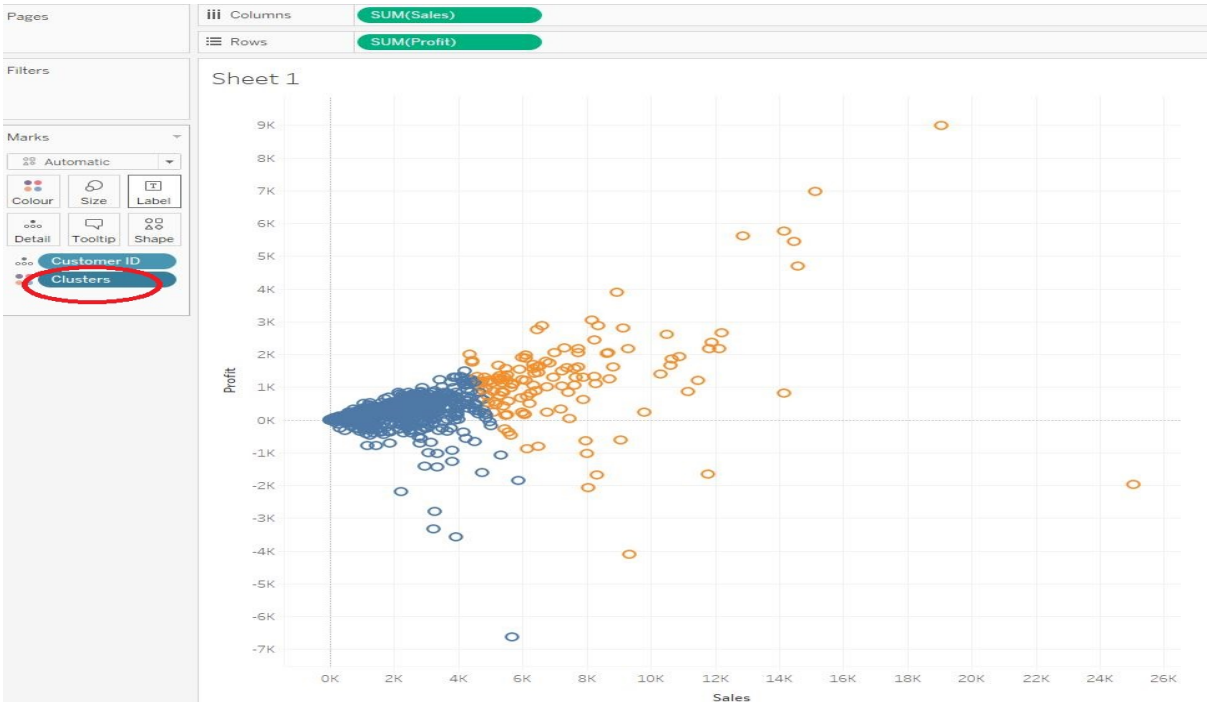
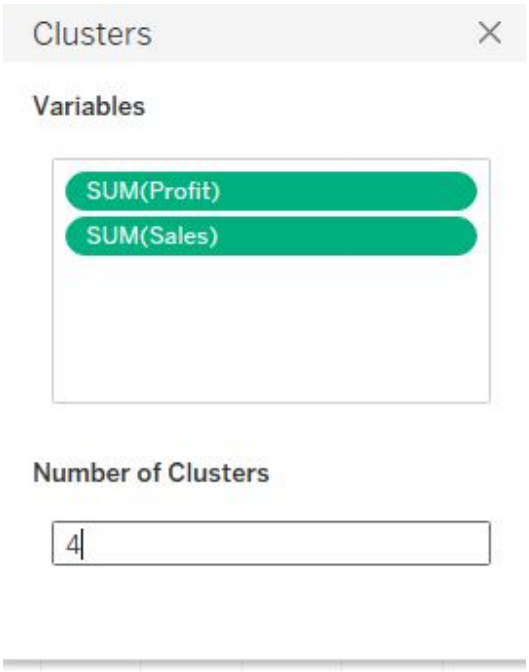


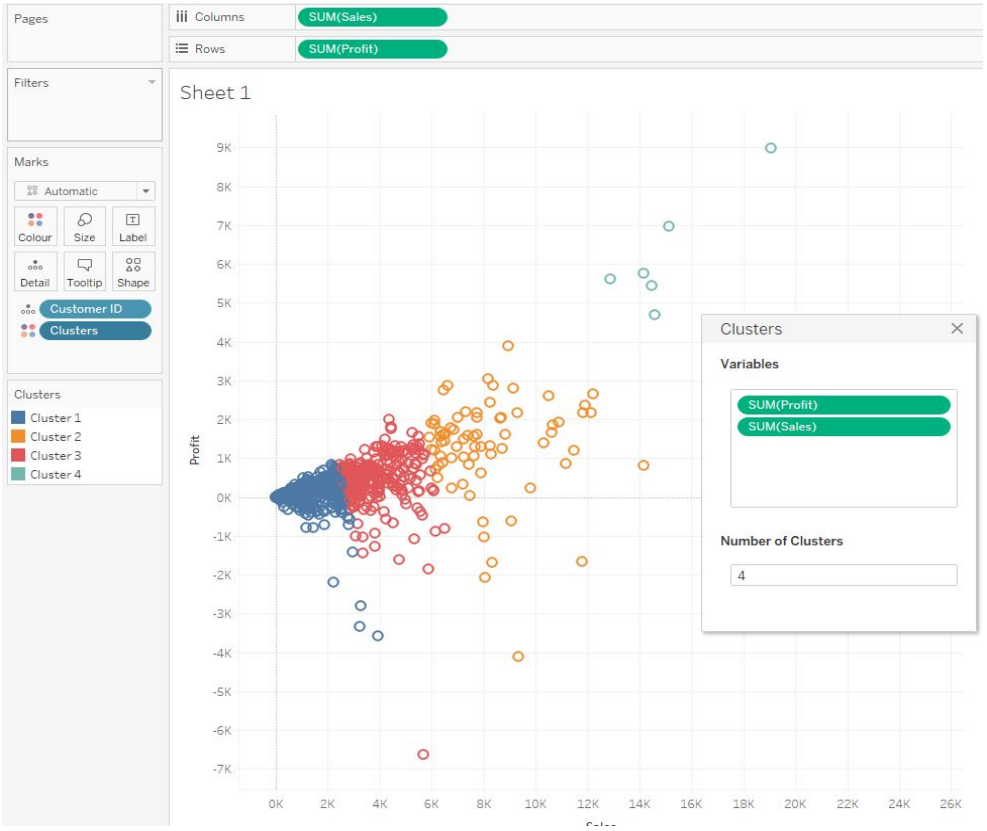

Tableau creates a new dimension called Clusters and places it on the Color field of the Marks card. Tableau automatically suggests a number of distinct clusters. In this case, it divides the marks into two segments (two colors).



In addition, a window opens that lets you make changes to the cluster analysis.

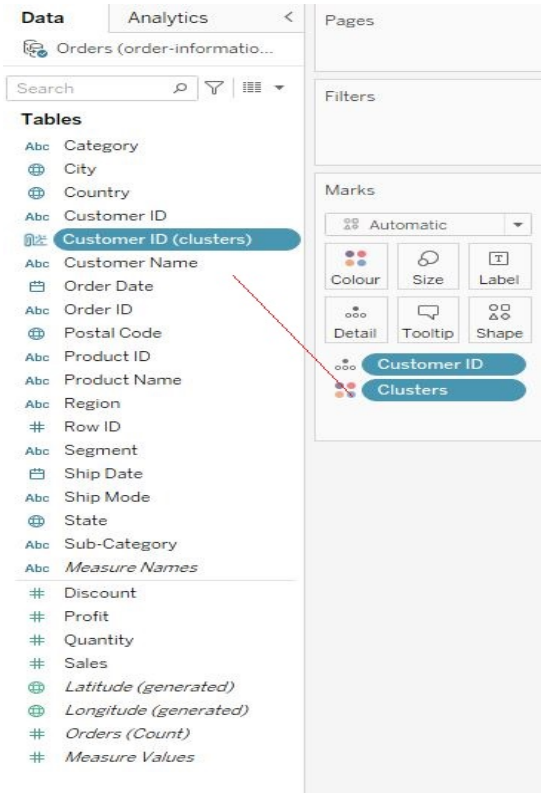



After entering 4 in that field, Tableau divides the marks into four segments:



**Step 4. Save clustering results**

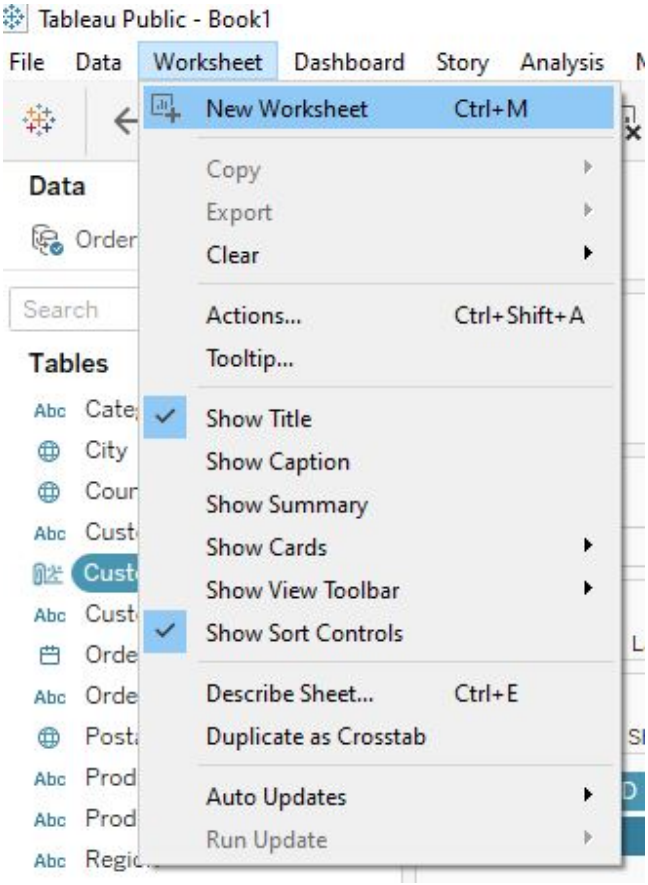
To save the results of a cluster analysis, you can drag the newly created Clusters field from the Marks card to the Data pane. Give the field a name or confirm the one suggested (in this case, Customer ID (clusters)). You can now find the field in the Dimensions section of the Data pane and use it like any other field.



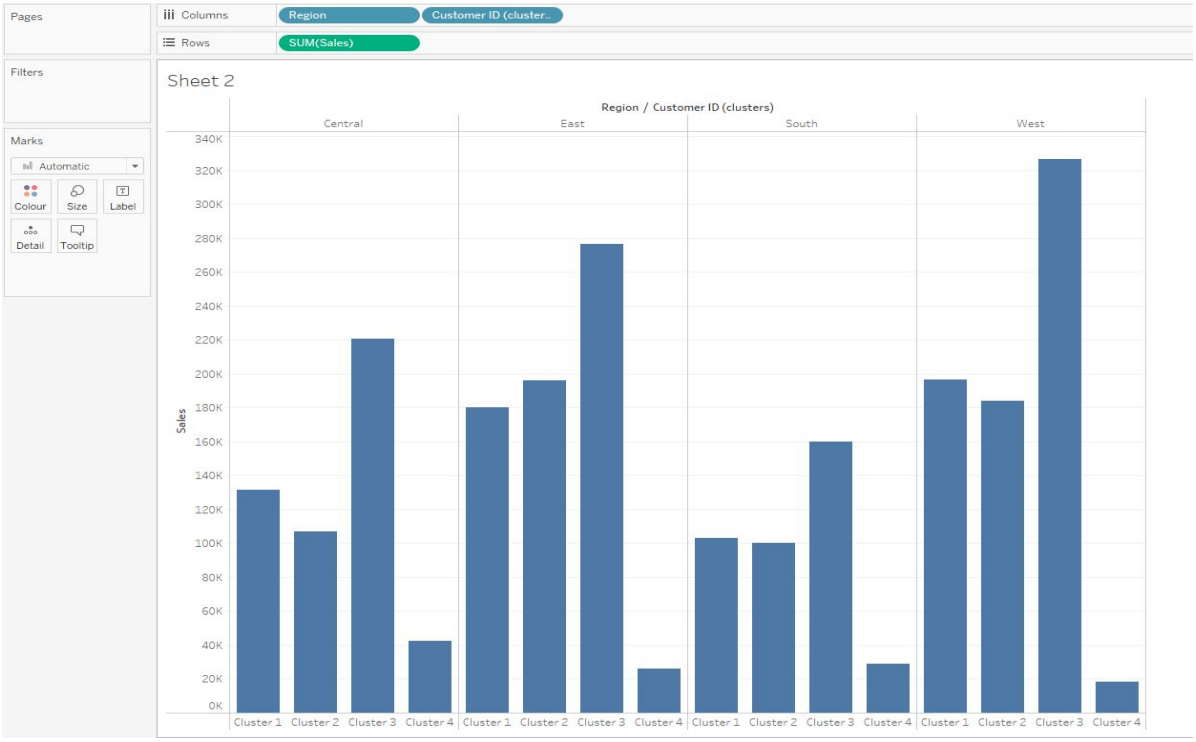

**Step 5. Work with clustering results**

You can use the newly created Clusters to see how the different cluster segments are distributed across the regions in the data set.

Open a new sheet from Worksheet in the same workbook




Then drag Sales onto Rows shelf and drag Region onto Columns shelf. Also pull the newly created clusters field Customer ID (clusters) onto Columns, as shown:




This is just a simple example of how you can continue to work with the results of a cluster analysis to ask further questions about your data.

## Project 4: Use parameter control

Parameters enable you to capture users input on your worksheets and dashboards, which is one of designing approaches for interactive data visualization. The captured values can then be used in calculated fields or be displayed as reference lines in your charts. Parameter-control elements can be added to worksheets and dashboards that allow the end user to either select a value from a predefined list or freely enter a value in a text field.

One potential use case for parameters is a what-if analysis.

**Task:** The given dataset provides information about order, including sales, profit, city, order ID, etc. The goal of this task is to check different sales growth scenarios for the given dataset and allow users to set the value for some controls.

**1. Download Dataset:** order-information.xlsx from Week 8 in Moodle

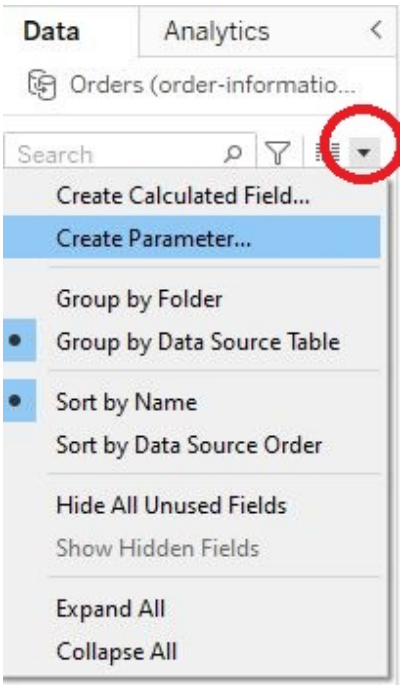
### Step 1. Load the data into Tableau

**Drag** Orders table to canvas and drop it and check that your data has been loaded correctly.

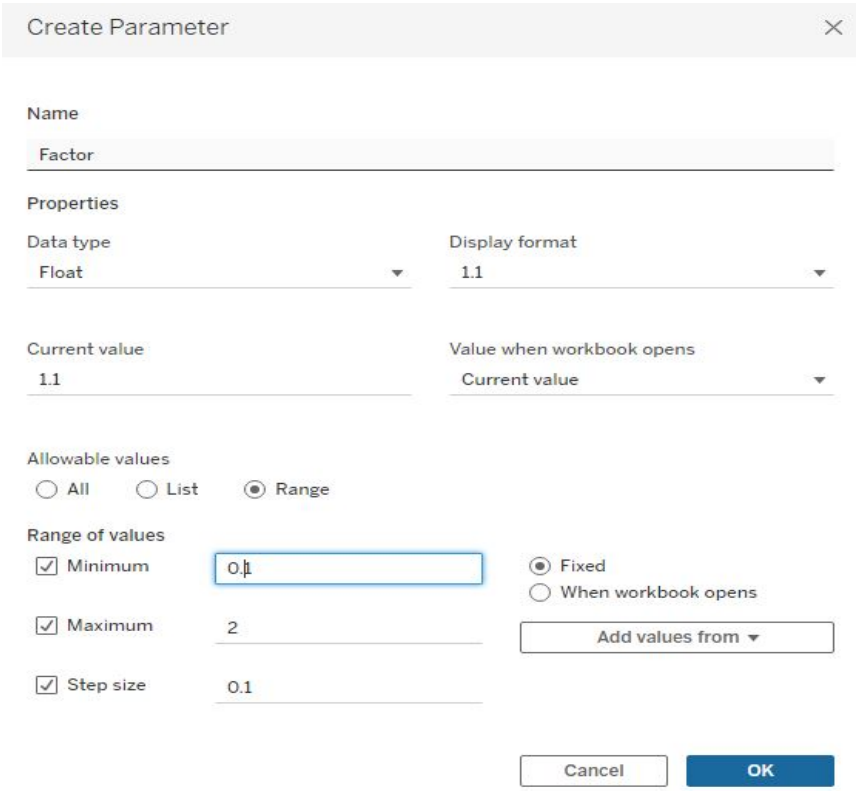
After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

### Step 2. Create a parameter

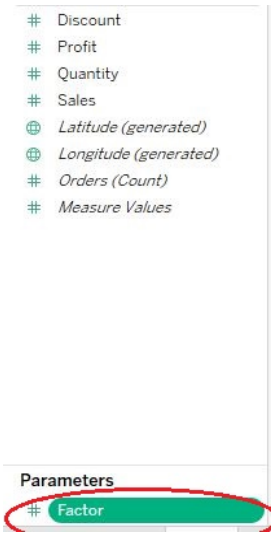
Now let us create a parameter. Click the small arrow at the top of the Data pane and select Create Parameter, as shown in




After clicking create parameter, a dialog box for configuring the new parameter appears. Let's call this parameter Factor. Set the data type to Float, define the current value as 1.1, and set the allowable values to Range. Also set the minimum value to 0.1, the maximum to 2, and the step size 0.1, as shown in

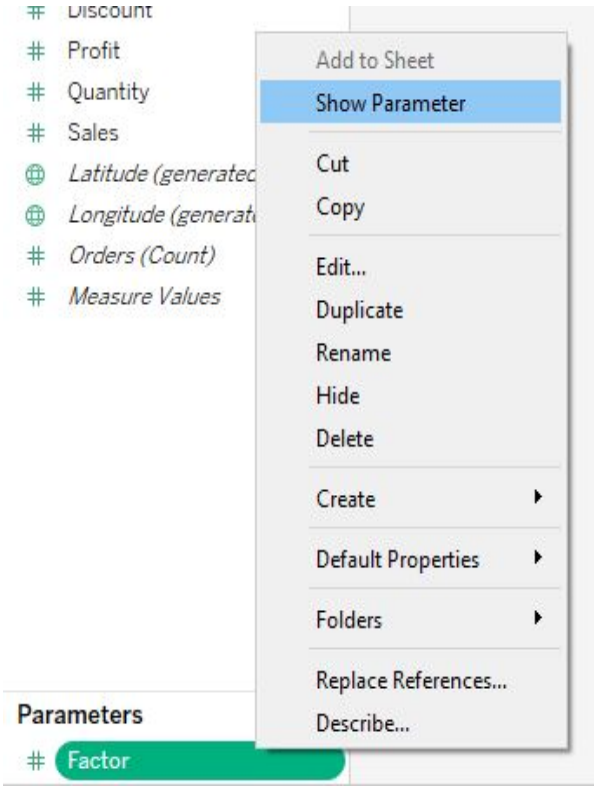



After clicking OK, the newly created parameter Factor will appear at the bottom of the Data pane in a new Parameters section.



**Step 3. Add parameter control element to the view**

To add the parameter's control element to the view, open the parameter's context menu (by right-clicking it), as shown in



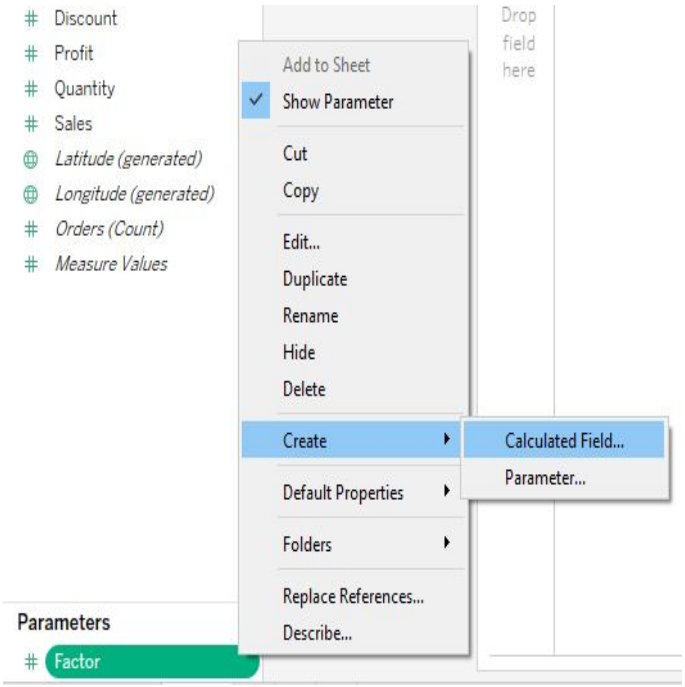

After selecting Show Parameter, the parameter's control element appears on the right side of the Tableau window: In this case, we get a slide control labeled Factor. Using this slider, the end user can now set the value for the factor, choosing from preset values between 0.1 and 2.



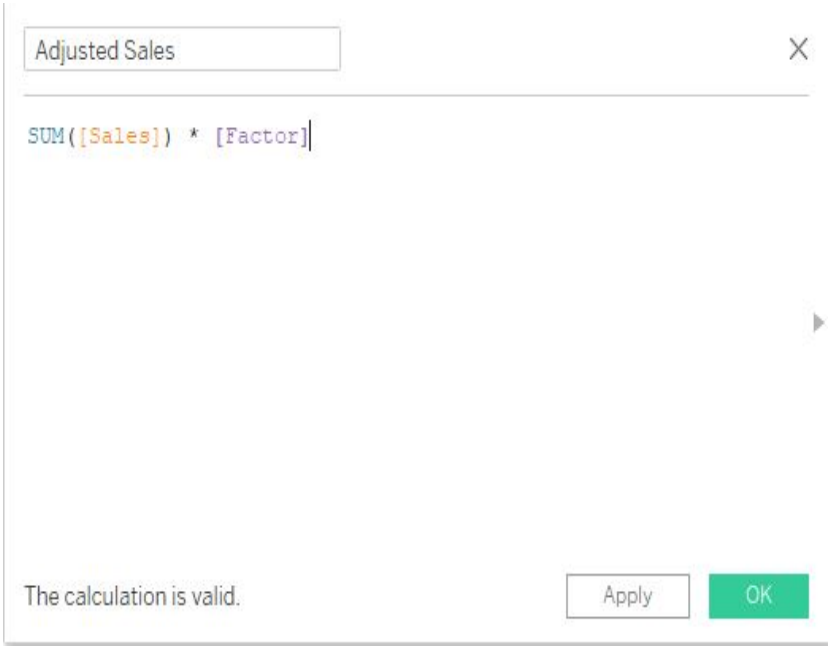
**Step 4. Apply parameter control (1)**

We now use the value set via the parameter control to calculate adjusted sales revenue figures. We want to solve the question what if revenue were to double?

To do so, create a new calculated field with Parameter Factor. Click Factor and select Create | Calculated Field.



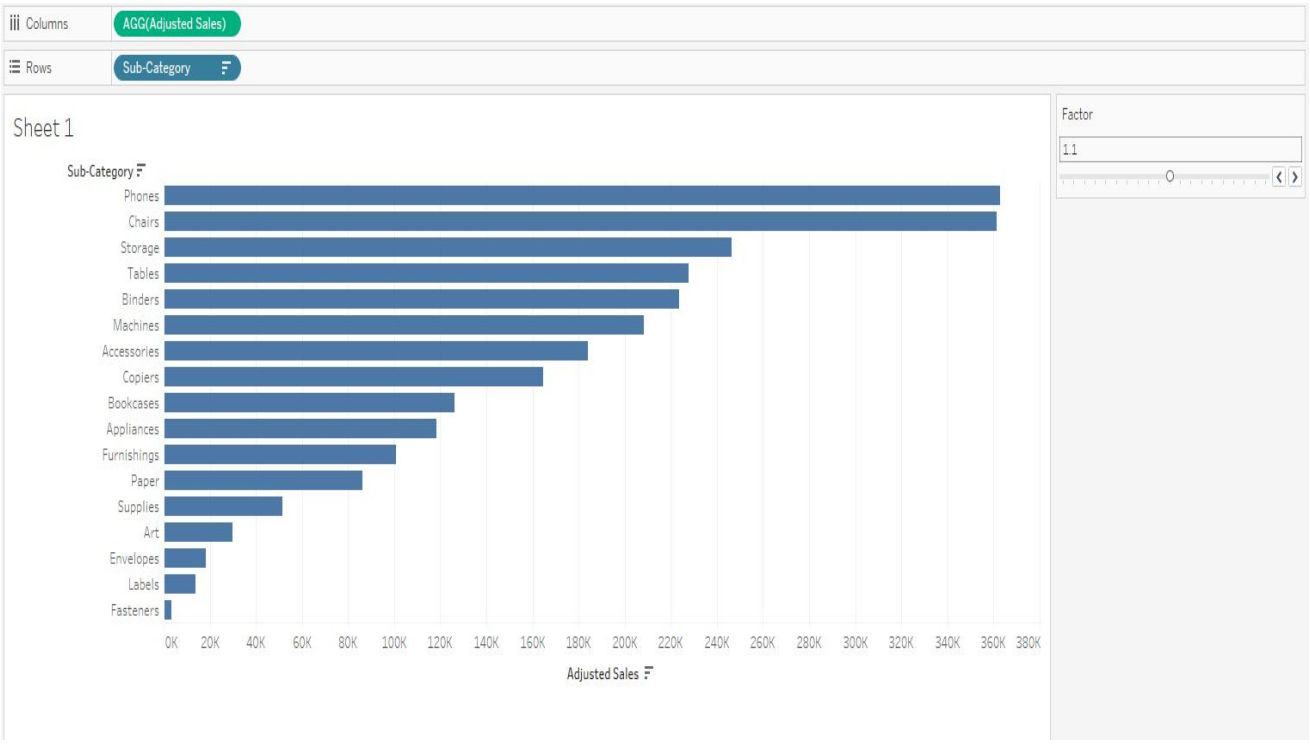

In the following editor box, enter the name Adjusted Sales, and enter the following formula:




The purple formatting of the Factor field in the editor of the calculated field is an indication that it is a parameter.

After clicking OK, new calculated field of Adjusted Sales you just created appears in the Measures area.

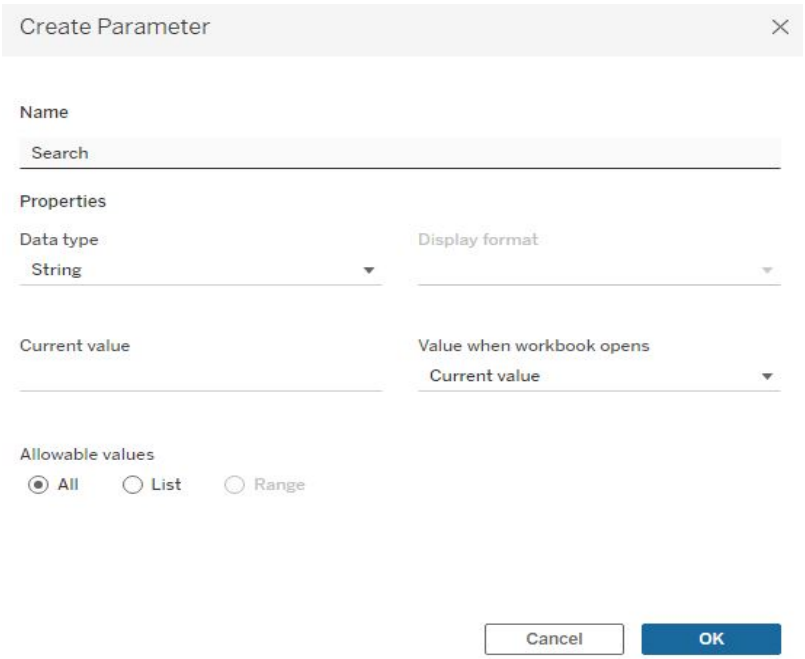
Drag Adjusted Sales from Data pane to Columns shelf, and drag the Sub-Category dimension to Rows shelf and sort the bars by their length:



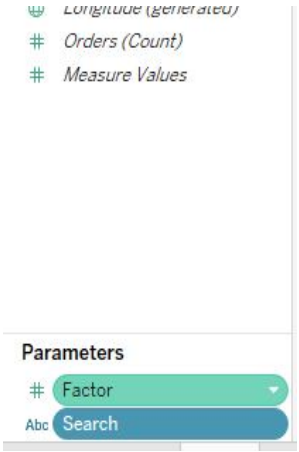
**Step 5. Apply parameter control (2)**

We now create a search box to look up customers by name, even if we don't know the whole name or the exact spelling.

Now let us create a parameter. Click the small arrow at the top of the Data pane and select Create Parameter, and create a new parameter called Search, set the data type to String, and leave the current value blank:




After clicking OK, the newly created parameter Search will appear at the bottom of the Data pane in a new Parameters section.



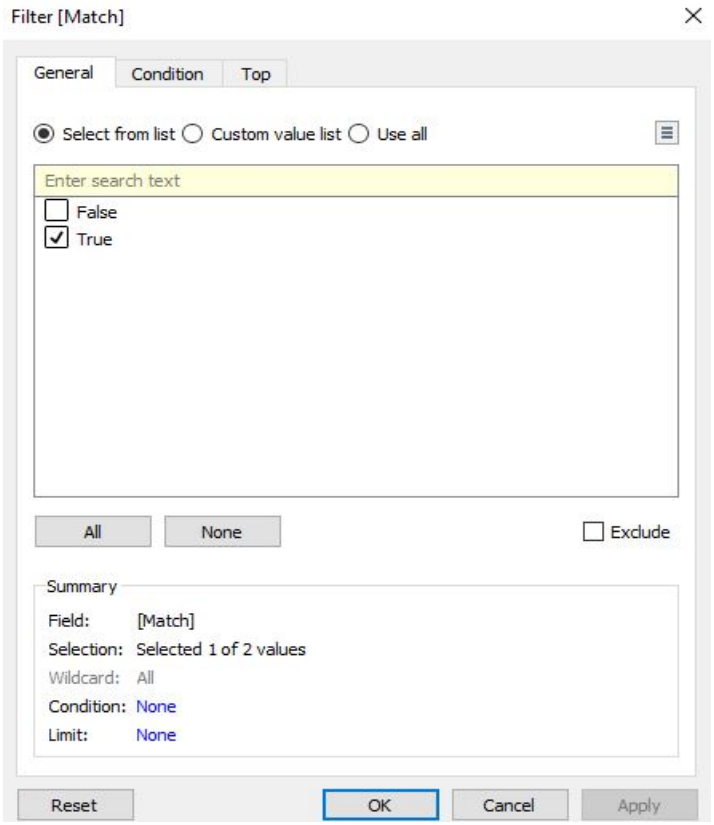
We then create a new calculated field to capture value in parameter Search. Click Search and select Create | Calculated Field.

In calculated field editor box, enter the name of the calculated field as Match, and enter the following formula:

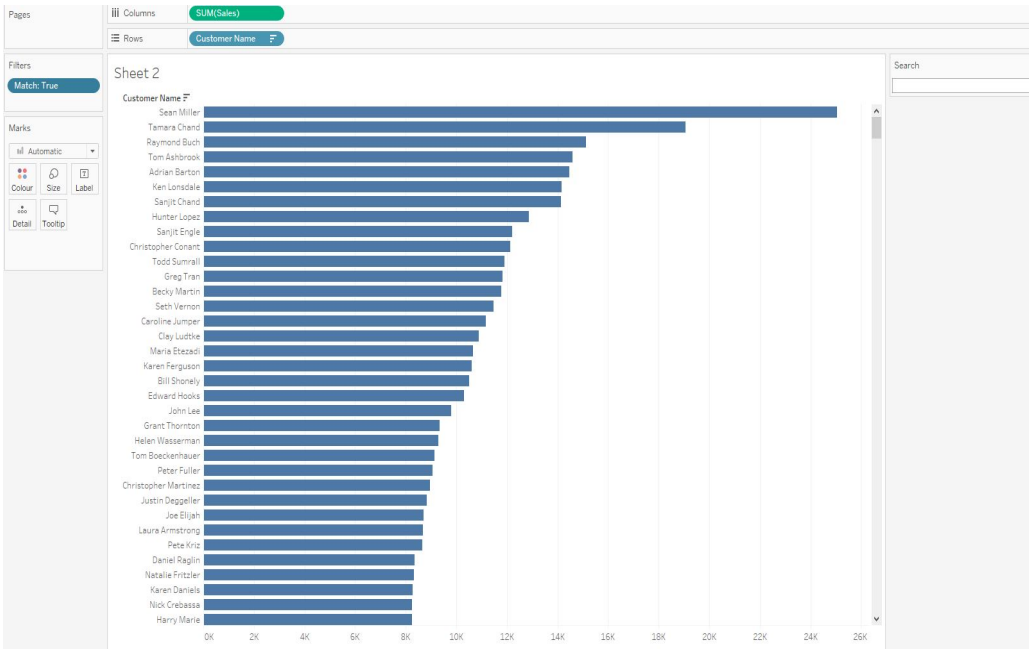



Here, we use the text operator CONTAINS to test whether the current value of the parameter is contained within the customer's name. To increase the hit rate, we convert both the customer's name and the search text to lowercase by using the LOWER function.

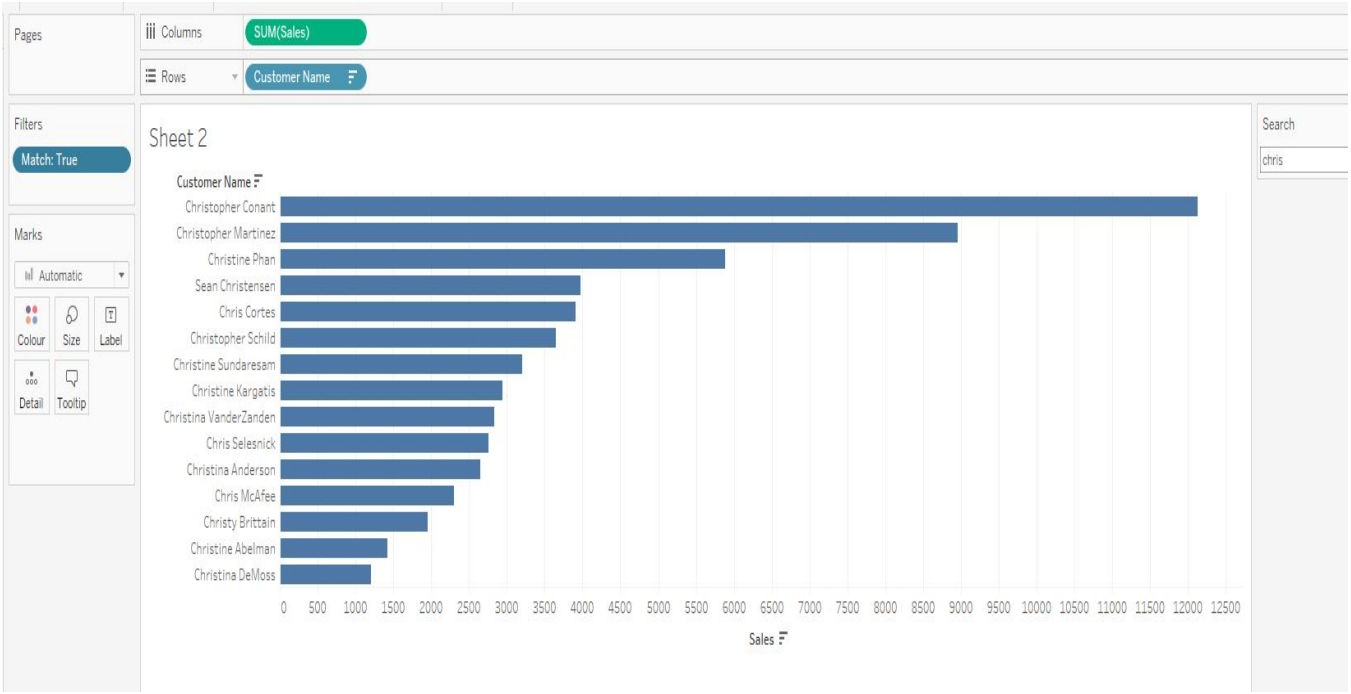
Next, drag the newly created Match field to the Filter card and choose True in the Filter dialog, as shown




Finally, create a new Worksheet—for example, by dragging Sales onto Columns shelf and drag Customer Name onto the Rows shelf. Since the search field (the parameter control on the right side) is still empty, you get the sales figures associated with all customers in the database.



We start to search by typing text (for instance, chris) into the search box, we now only get sales data for customers whose name contains the character string chris.



## Project 5. Create a Professional Report

Summarize the above experiments procedure, results, answering for questions and screenshots (project 1, 2, 3, 4 ) into one report. Your report is the assignment that is required to be submitted for evaluation on week 8. Create a report by following the below steps.

Please note that

- as you may have a different version of Microsoft Word, the **following steps are just for your references.**

You can add a chapter called Chapter 8 in your previous report.

1. Open your last week's report and find the end of last week's report.
2. Copy this week's related experimental results, your findings and Screenshots, and paste them at the end of last week's report.
3. Delete original Table of Content you created.
4. Select all content , align all text to both left and right margin
5. Use shortcut key approach, generate Chapter 8: **Complex data visualizations (3)**
6. Then use shortcut key approach to generate proper sub-chapters for this week's lab work.
7. Insert Table of Contents to your report.